WEEK-8 TASK

Macro-Economic Causal Modeling with Vector Autoregression (VAR) Analytics

1. Introduction

Macroeconomic indicators such as Gross Domestic Product (GDP), inflation, and interest rates play an important role in understanding a country's economic performance. These indicators are interconnected and can influence one another over time.

Vector Autoregression (VAR) is a statistical model used to analyze the relationships between multiple time-series variables. It helps identify how changes in one economic indicator affect other indicators in the future.

This project uses VAR analytics to study macroeconomic relationships, identify suitable time lags, and understand the effects of economic changes through impulse-response analysis.

2. Objectives

- To analyze historical GDP, inflation, and interest-rate data.
- To test the stationarity of economic time-series data.
- To identify the optimal lag using the Akaike Information Criterion (AIC).
- To build and estimate a VAR model.
- To perform impulse-response analysis.
- To understand economic relationships and forecast future trends.

3. Problem Statement

Economic indicators frequently change due to government policies, market conditions, inflation, and other economic factors. These changes can affect business investment, production, borrowing costs, and market performance.

Therefore, a statistical model is needed to analyze the interactions between GDP, inflation, and interest rates. The VAR model helps examine these relationships and understand how economic shocks may influence future economic conditions.

4. Tools and Technologies

The following tools are used in this project:

- **Python:** For data processing and analysis.
- **Pandas:** For handling economic datasets.
- **NumPy:** For numerical calculations.
- **Matplotlib:** For creating graphs.
- **Statsmodels:** For stationarity testing and VAR modeling.
- **Jupyter Notebook or Google Colab:** For executing the Python program.

5. Dataset Description

The project uses historical macroeconomic data containing the following variables:

Variable	Description
Year	Year of observation
GDP Growth	Annual percentage growth in GDP
Inflation	Annual percentage change in the general price level
Interest Rate	Annual interest rate expressed as a percentage

The dataset should contain observations over multiple years to identify long-term economic patterns. Actual historical data from a reliable source should be used for meaningful analysis.

6. Methodology

The project follows these steps:

Step 1: Data Collection

Collect historical GDP growth, inflation, and interest-rate data from reliable economic databases.

Step 2: Data Preprocessing

Check for missing values, arrange the observations chronologically, and select the required variables.

Step 3: Stationarity Testing

Use the Augmented Dickey-Fuller (ADF) test to determine whether the time-series variables are stationary.

Step 4: Lag Selection

Use the Akaike Information Criterion (AIC) to select an appropriate lag order for the VAR model.

Step 5: VAR Model Estimation

Fit the VAR model using the stationary data and the selected lag order.

Step 6: Impulse-Response Analysis

Analyze how a shock to one economic variable affects the other variables over time.

Step 7: Forecasting

Generate forecasts for the selected economic indicators using the fitted VAR model.

7. Mathematical Concepts

A. Stationarity

A time series is stationary when its statistical properties, such as mean, variance, and autocovariance, remain stable over time.

The Augmented Dickey-Fuller test evaluates the null hypothesis that a series has a unit root.

- If the p-value is less than 0.05, reject the null hypothesis and treat the series as stationary under the test.
- If the p-value is greater than or equal to 0.05, there is insufficient evidence to reject the unit-root hypothesis.

If a series is non-stationary, differencing or another appropriate transformation may be required.

B. Akaike Information Criterion (AIC)

AIC is used to compare statistical models and select a suitable lag order.

AIC is calculated as:

$AIC = 2k - 2\ln(L)$

Where:

- k = Number of estimated parameters.
- L = Maximum likelihood of the model.

A lower AIC generally indicates a better balance between model fit and complexity among the candidate models.

C. Vector Autoregression (VAR)

A VAR model explains each variable using its own past values and the past values of the other variables.

For a three-variable system:

$Y_t = c + A_1Y_{t-1} + A_2Y_{t-2} + \ldots + A_pY_{t-p} + \varepsilon_t$

Where:

- Y_t = Vector containing GDP growth, inflation, and interest rate.
- c = Constant vector.
- $A_1, A_2, \ldots, A_p$ = Coefficient matrices.
- p = Selected lag order.
- ε_t = Error or innovation vector.

D. Impulse-Response Analysis

Impulse-response analysis measures how a shock to one variable affects the current and future values of variables in the VAR system.

For example, it can help analyze how an unexpected increase in interest rates may affect GDP growth and inflation over subsequent periods.

8. Python Implementation

The following program demonstrates the complete VAR workflow using a sample dataset. The sample values are illustrative and are not actual historical economic observations. Replace them with real economic data before drawing conclusions about the economy.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.api import VAR

# 1. Create illustrative sample data
np.random.seed(42)
years = np.arange(2000, 2025)
n = len(years)

data = pd.DataFrame({
   "GDP_Growth": np.random.normal(5.5, 1.0, n),
   "Inflation": np.random.normal(4.5, 1.2, n),
   "Interest_Rate": np.random.normal(6.0, 0.8, n)
}, index=years)

data.index.name = "Year"
print("First five observations:")
print(data.head())

# 2. Plot the economic indicators
data.plot(subplots=True, figsize=(10, 7))
plt.suptitle("Illustrative Macroeconomic Indicators")
plt.tight_layout()
plt.show()
```

```
# 3. Test stationarity using the ADF test
print("\nADF Stationarity Test")

for column in data.columns:
    result = adfuller(data[column])

    print(f"\n{column}")
    print("ADF Statistic:", round(result[0], 4))
    print("p-value:", round(result[1], 4))

# 4. Select lag order using AIC
model = VAR(data)

lag_selection = model.select_order(maxlags=4)
print("\nLag Selection Results:")
print(lag_selection.summary())

# Select the lag with the lowest AIC
selected_lag = lag_selection.selected_orders["aic"]

if selected_lag is None or selected_lag < 1:
    selected_lag = 1

print("\nSelected Lag:", selected_lag)

# 5. Fit the VAR model
var_model = model.fit(selected_lag)

print("\nVAR Model Summary:")
print(var_model.summary())
```

```
# 6. Check model stability
print("\nIs the VAR model stable?", var_model.is_stable())

# 7. Impulse-response analysis
irf = var_model.irf(10)

irf.plot(orth=False)
plt.suptitle("Impulse-Response Analysis")
plt.tight_layout()
plt.show()

# 8. Forecast the next five periods
forecast_steps = 5

last_observations = data.values[-selected_lag:]

forecast = var_model.forecast(
    y=last_observations,
    steps=forecast_steps
)

forecast_years = np.arange(
    data.index[-1] + 1,
    data.index[-1] + forecast_steps + 1
)
forecast_df = pd.DataFrame(
    forecast,
    index=forecast_years,
    columns=data.columns
)
```

```
print("\nIllustrative Five-Period Forecast:")
print(forecast_df.round(3))

# 9. Plot forecasts
forecast_df.plot(
    subplots=True,
    figsize=(10, 7),
    marker="o"
)

plt.suptitle("Illustrative VAR Forecast")
plt.tight_layout()
plt.show()
```

9. Expected Analysis and Results

After running the program with actual economic data, the following results should be examined:

A. Stationarity Results

The ADF test provides a test statistic and p-value for each variable. These results help determine whether differencing or another transformation is necessary.

B. Optimal Lag Selection

The AIC output identifies a preferred lag order among the candidate models. The selected lag determines how many previous periods are included in the model.

C. VAR Model Results

The model summary provides estimated coefficients, standard errors, test statistics, and other information. These values help explain how past economic indicators are associated with current values.

D. Impulse-Response Results

The impulse-response graphs show the direction, size, and duration of responses to shocks within the model. Confidence intervals should be considered when interpreting the responses.

E. Forecasting Results

The model generates forecasts for GDP growth, inflation, and interest rates for future periods. Forecast accuracy should be evaluated using suitable out-of-sample measures.

10. Business Applications

VAR analytics can support business and economic decision-making in several ways:

1. **Investment Planning:** Helps analysts understand relationships between economic growth and interest rates.
2. **Risk Management:** Helps businesses examine how economic shocks may affect their operations.
3. **Market Analysis:** Provides insights into economic conditions associated with market movements.
4. **Financial Planning:** Supports the assessment of borrowing costs and inflation trends.
5. **Strategic Decision-Making:** Helps businesses evaluate possible economic scenarios.

However, macroeconomic VAR forecasts alone cannot reliably predict stock prices or guarantee market movements.

11. Advantages

- Analyzes multiple economic variables simultaneously.
- Captures relationships across different time lags.
- Supports systematic forecasting.
- Helps study the effects of economic shocks.
- Provides useful information for economic and business planning.

12. Limitations

- Requires sufficient and reliable historical data.
- Results depend on model specification and lag selection.
- Non-stationary data can produce misleading results if handled incorrectly.
- Forecast accuracy may decline when economic conditions change.
- Impulse-response results depend on how shocks are identified.
- Statistical relationships do not automatically establish true causation.

13. Conclusion

The Macro-Economic Causal Modeling project demonstrates how Vector Autoregression can be used to analyze relationships between GDP growth, inflation, and interest rates. Stationarity testing helps assess the time-series properties of the data, while AIC supports lag selection. The VAR model estimates relationships among the selected variables, and impulse-response analysis examines how shocks may influence the economic system over time.